

Load Interconnection System Impact Study Report

For Queue Project Q20-1381

Lewistown 138 kV

April 25, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1381. The Interconnection Customer proposes a 79.2 MW large load (data center) facility interconnecting at the Lewistown 138 kV substation in the Pennsylvania area. The request was received on January 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 473 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$11,946,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Rewey Substation 345 kV/138 kV Auto	Ft Humbug / Orchard Road 500kV	149.0	446	13.0
Honk Falls / Loco Hills 138kV line	Loco Hills / Rewey 138kV L	111.0	172	30.3
Naples / Puna Geothermal Venture I 138kV line	Edes Tap 2 / Lutesville 345kV	107.4	256	5.7
Loco Hills / Lewistown 138kV	W Campus / Fort Humbug 345kV	107.3	428	41.1
Silverton / Baja 138kV Line	Edes Tap 2 / Lutesville 345kV	105.2	247	5.7
Sfpp Lp / Grinding 138kV	Edes Tap 2 / W Campus 345kV	104.7	320	8.0
Strahan / Grinding 138kV	West Campus Substation / Edes Tap 2 345kV	104.5	324	8.0
Marion S / Puna Geothermal Venture I 138kV	Edes Tap 2 / Lutesville 345kV	104.1	266	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rewey Substation 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$15,513,000	10% (\$1,567,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Honk Falls / Loco Hills 138kV line	Rebuild line with higher-capacity conductor	\$7,460,000	73% (\$5,431,000)
Naples / Puna Geothermal Venture I 138kV L	Reconductor line	\$2,588,000	20% (\$529,000)
Loco Hills / Lewistown 138kV	Reconductor line	\$1,608,000	48% (\$765,000)
Silverton / Baja 138kV Line	Reconductor line	\$7,912,000	20% (\$1,617,000)
Sfpp Lp / Grinding 138kV	Reconductor line	\$1,971,000	15% (\$289,000)
Strahan / Grinding 138kV	Reconductor line	\$1,204,000	15% (\$176,000)
Marion S / Puna Geothermal Venture I 138kV	Reconductor line	\$7,694,000	20% (\$1,572,000)
	Total allocated to project		\$11,946,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.